

MUHAMMAD HUZAIFA

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About

Full Stack Engineer with 3+ years architecting and shipping production systems across FinTech and DefTech; from a multi-region payment gateway processing thousands of daily transactions to secure embedded systems for critical infrastructure. Fluent across Java(Spring Boot) and C#/.NET microservices, React, and cloud infrastructure (Azure, Kubernetes, CI/CD).

Technical Skills

- **Languages:** Java, C#, Python, Go, JavaScript, TypeScript, SQL, C, C++
- **Backend & Frameworks:** Spring Boot, ASP.NET Core, Node.js, REST APIs, API Design, Microservices, Hibernate
- **Frontend:** React JS, TypeScript
- **Databases:** PostgreSQL, MSSQL, MySQL, YugabyteDB
- **Messaging & Event-Driven Systems:** Apache Kafka, RabbitMQ
- **Cloud & DevOps:** Azure (VNet, AKS, Key Vault, VMs), Google Cloud, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD
- **Software Engineering:** System Design, Distributed Systems, Object-Oriented Programming, Agile/Scrum, Unit Testing, Test Automation, Technical Documentation
- **Security & Compliance:** Payment Card Industry Data Security Standard (PCI-DSS), GDPR, OWASP Top 10, SIEM, Vulnerability Assessment, Secure System Design
- **AI Tools:** Claude Code, Codex, Cursor, GitHub Copilot

Experience

Research Intern

June 2026– Present

University of Oulu

Oulu, Finland

- Extended the open-source AVISE security evaluation framework to Text-to-Image generative models, designing adversarial and prompt-injection resistance tests across 5 model providers, including Nano Banana Pro (Gemini 3 Pro Image), Grok Imagine, and GPT-5.4 Image 2.
- Co-authoring a research paper formalizing security vulnerability methodologies for generative AI systems, using Python, Docker, and OpenRouter.
- Built OpenRouter-based model connectors enabling standardized, reproducible security evaluation across 4+ T2I model providers spanning Google, X-AI, OpenAI, and open-source diffusion architectures.

Full Stack Developer

February 2024– March 2026

Datapulse Technologies (FinTech)

Ireland (Remote)

- Architected and delivered a full replacement payment gateway platform end-to-end, closing critical functional gaps in a legacy multi-region system used in production.
- Improved system performance by 25% and reduced latency on a legacy whitelabel payment architecture, sustaining stability under peak transaction loads.
- Integrated 3+ global acquirers (E-Comprocessing, Transflo, Rapyd) and digital wallets (Apple Pay, Google Pay) via REST APIs, enabling secure multi-region payment processing.
- Unified fragmented physical terminal systems into a single gateway supporting both card-present and card-not-present transactions.
- Built and scaled a microservices architecture (Java/Spring Boot) processing thousands of daily transactions with high fault tolerance.
- Owned CI/CD pipeline design (GitHub → Jenkins → Docker → Kubernetes), enabling zero-downtime, single-push production releases.
- Led PCI-DSS compliance readiness end-to-end, implementing secure data handling, access controls, and SIEM/Cloud Defender audit logging —closing 15+ audit findings ahead of certification.

Trainee Engineer

July 2022– May 2024

Rapidev (DefTech)

Islamabad, Pakistan

- Integrated 3 hardware sensors into the RapiGuard Anti-UAV system, enabling real-time aerial threat detection.
- Built a Key Management Application handling full encryption key lifecycles for critical military communications.
- Engineered secure cryptographic key injection via UART, ensuring protected hardware-software data transfer for defense-grade systems.

Projects

Nafalytics

Founder

Stock Portfolio Tracker

- Built and launched a portfolio tracking platform from scratch in 3 months, growing to 300 active users tracking real-time P/L and trade activity.
- Developed real-time profit/loss tracking, historical performance analytics, automated payouts, and a notification engine.
- Owned full product lifecycle and cloud deployment (Java Spring Boot, React JS, PostgreSQL, Azure) with automated CI/CD via GitHub Actions.
- Deployed here: <https://nafalytics.com>

Education

MSc Computer Science & Engineering

September 2025 – June 2027

University of Oulu

Oulu, Finland

BSc Computer Science

August 2020– May 2024

National University of Sciences & Technology

Islamabad, Pakistan

- **Publication:** OptiGuard: Generalized, Attention-Driven & Explainable Glaucoma Classification
 - * Achieved state of the art results on G-1020 & SMDG datasets.
 - * Awarded 1st Best Project distinction.
 - * Project details: <https://linktr.ee/optiguard>

Certifications

Junior Penetration Tester

December 2025

TryHackMe

Course Work

- Hands-on experience with OWASP Top 10 vulnerabilities including SQL Injection (SQLi), Cross-Site Scripting (XSS), IDOR, and Command Injection.
- Skilled in Authentication Bypass, Server-Side Request Forgery (SSRF), and File Inclusion (LFI/RFI) exploitation and remediation.